

# MATHEMATICAL STATISTICS

## 1. BASIC SETUP

**Definition 1.1** (Statistical setup). A *statistical setup* consists of the following data:

We have the following measurable spaces:

- The *observation space*  $\mathcal{X}$ .
- The *parameter space*  $\Theta$ .
- The *action space*  $\mathcal{A}$ .
- The *sample space*  $\Omega$

We also have the following maps:

- A *loss function*  $L : \Theta \times \mathcal{A} \rightarrow \mathbb{R}$ .
- An *observation*  $X : \Omega \rightarrow \mathcal{X}$ .
- A *decision function*  $d : \mathcal{X} \rightarrow \mathcal{A}$ .

For each parameter  $\theta \in \Theta$ , we have a probability measure  $P_\theta \in \text{Prob}(\Omega)$ . The associated risk function is then

$$\begin{aligned} R(\theta, d) &= E_\theta L(\theta, d(X)) \\ &= \int_{\Omega} L(\theta, d(X(\omega))) dP_\theta(\omega). \end{aligned}$$

Note: It seems that unless otherwise specified, it does not hurt to just assume  $\mathcal{X} = \Omega$  and  $X$  is the identity map in which case

$$R(\theta, d) = \int_{\mathcal{X}} L(\theta, d(x)) dP_\theta(x).$$

**Interpretation:** We have an unknown parameter  $\theta$  and a given observation  $X(\omega) \in \mathcal{X}$  where  $\omega \in \Omega$  was randomly sampled according to  $P_\theta$ . Based on this observation, we take an action  $d(x) \in \mathcal{A}$ . Then the loss  $L(\theta, d(x))$  measures how wrong we are in taking this action  $d(x)$ . The risk  $R(\theta, d)$  is then the average of this loss (for the fixed  $\theta$ ).

**Example 1.2.** Suppose now  $\Omega = \mathcal{X} = \mathbb{R}^n$  and  $X$  is identity (as usual) and  $\mathcal{A} = \mathbb{R}$ . Let  $\Theta = \mathbb{R} \times \mathbb{R}_{\geq 0}$  and for  $\theta = (\mu, \sigma^2) \in \Theta$  we let  $P_\theta$  denote the distribution of the random variable  $(X_1, \dots, X_n) \in \mathbb{R}^n$  where the  $X_i$  are i.i.d  $N(\mu, \sigma^2)$ . We take a loss function  $L(\theta, a) = |\mu - a|^2$ . Thus if  $d : \mathcal{X} \rightarrow \mathcal{A}$  is a decision rule then  $L(\theta, d(x))$  will measure how well of an approximation  $d(x) \in \mathbb{R}$  is of the unknown  $\mu$ . Let us choose  $d(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i$ . Thus

$$R(\theta, d) = E \left| \frac{1}{n} \sum_{i=1}^n X_i - \mu \right|^2 = \text{Var} \left( \frac{1}{n} (X_1 + \dots + X_n) \right) = \frac{\sigma^2}{n}$$

by the independence of the  $X_i$ . We see that as  $R(\theta, d) = O(n^{-1})$  so this seems to be a good decision rule.

If instead we use the loss function  $L(\theta, a) = |\sigma^2 - a|$  then a good decision rule seems to be

$$d(X) = s^2 := \frac{1}{n-1} \sum_{i=1}^n |X_i - \bar{X}|^2.$$

It turns out that (TODO?)

$$R(\theta, d) = \frac{2\sigma^4}{n-1}.$$

We can use this language to define hypothesis testing:

**Definition 1.3.** Suppose now that we have a partition  $\Theta = \Theta_0 \sqcup \Theta_1$  of the parameter space and our action space is  $\mathcal{A} = \{\Theta_0, \Theta_1\}$ . We have a 0 – 1 loss function given by

$$L(\theta, \Theta_i) = 1 - \mathbf{1}_{\Theta_i}(\theta).$$

Thus  $L(\theta, a)$  measures whether we chose the right partition  $a \in \mathcal{A}$  that  $\theta$  landed in. We see that the risk

$$R(\theta, d) = P_\theta(\{\omega \in \Omega \mid \theta \notin d(X(\omega))\})$$

is the probability that our decision algorithm chooses the wrong partition.

The action  $\Theta_0$  is called the *Null Hypothesis* while  $\Theta_1$  is called the *Alternative hypothesis*. A *Type I* error is an action that chooses the alternative hypothesis ( $\Theta_1$ ) when the null hypothesis is true ( $\theta \in \Theta_0$ ). A *Type II error* is an action that chooses the null hypothesis when the alternative is true.

**Example 1.4.** Suppose we have a dice that is either fair (null hypothesis) or always rolls a 6 (alternative hypothesis). Suppose we have rolled this dice  $n$  times and observed the outcomes  $(X_1, \dots, X_n) \in \{1, 2, 3, 4, 5, 6\}^n = \mathcal{X}$ . Say  $\Theta = \{\theta_0, \theta_1\}$  where  $P_{\theta_0}$  is the uniform distribution on  $\mathcal{X}$  (dice is fair) while  $P_{\theta_1}$  is the probability measure concentrated on  $(6, \dots, 6) \in \mathcal{X}$ . Our decision rule simply chooses  $\theta_1$  if and only if we roll all 6. We have  $R(\theta_0, d) = 1/6^n$  while  $R(\theta_1, d) = 0$ .

**Definition 1.5.** We say that a statistical setup is *nice* if there exists a  $\sigma$ -finite positive measure (not necessarily a probability measure)  $\nu$  on  $\Omega = \mathcal{X}$  and a measurable function  $f : \mathcal{X} \times \Theta \rightarrow \mathbb{R}_{\geq 0}$  such that  $P_\theta \ll \nu$  and

$$\frac{dP_\theta}{d\nu}(x) = f(x|\theta) := f(x, \theta)$$

for all  $\theta \in \Theta$ .

For example, if  $\mathcal{X}$  is countable and thus each  $P_\theta$  is discrete, then  $\nu$  can be the counting measure.

**Assumption going forward:** We always assume a statistical setup is nice without mentioning it. This is the case for the examples above (i.e.,  $\nu$  is either a Lebesgue measure or a counting measure).

## 2. BAYESIAN SETUP

**Definition 2.1.** A *prior distribution* is a probability measure  $\mu_\Theta$  on the parameter space  $\Theta$ .

Recalling that we assume the nice condition of Definition 1.5, we thus have a probability measure  $P$  on  $\mathcal{X} \times \Theta$  given by

$$\frac{dP}{d(\nu \times \mu_\Theta)}(x, \theta) = f(x|\theta).$$

In other words for integrable  $\psi : \mathcal{X} \times \Theta \rightarrow \mathbb{R}$  we have

$$\begin{aligned}
\int_{\mathcal{X} \times \Theta} \psi(x, \theta) dP(x, \theta) &= \int_{\mathcal{X} \times \Theta} \psi(x, \theta) f(x|\theta) d(\nu \times \mu_\Theta)(x, \theta) \\
&= \int_{\Theta} \left( \int_{\mathcal{X}} \psi(x, \theta) f(x|\theta) d\nu(x) \right) d\mu_\Theta(\theta) \\
&= \int_{\Theta} \int_{\mathcal{X}} \psi(x, \theta) dP_\theta(x) d\mu_\Theta(\theta)
\end{aligned}$$

From now on, we let  $X : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$  and  $\Theta : \mathcal{X} \times \Theta \rightarrow \Theta$  denote the projections (note: by abuse of notation  $\Theta$  is now either a space or a projection map, but it should cause no confusion in any context).

Let  $P_X = X^*P$  denote the push-forward measure of  $P$  onto  $\mathcal{X}$ , that is,  $P_X(A) = P(A \times \Theta)$  for  $A \subset \mathcal{X}$ . We let

$$g(x) := \int_{\Theta} f(x, \theta) d\mu_\Theta(\theta).$$

Observe that for  $A \subset \mathcal{X}$  we have that

$$P_X(A) = \int_{A \times \Theta} f(x|\theta) d\mu_\Theta(\theta) d\nu(x) = \int_A g(x) d\nu(x)$$

and thus  $P_X$  has a density with respect to  $\nu$  given by  $g$ .

**Lemma 2.2.** For  $P_X$ -almost all  $x \in \mathcal{X}$  we have that

$$g(x) > 0.$$

*Proof.* Let  $A = \{x \in \mathcal{X} \mid g(x) = 0\}$ . Then

$$P_X(A) = \int_A g(x) d\nu(x) = 0.$$

□

Given this lemma, we can now define for  $P_X$ -almost all  $x \in \mathcal{X}$  a probability measure on  $\Theta$  given by

$$P_{\Theta|X=x}(B) = \frac{1}{g(x)} \int_B f(x|\theta) d\mu_\Theta(\theta)$$

for  $B \subset \Theta$ . This is known as the *posterior distribution of  $\Theta$  given  $X = x$* .

**Proposition 2.3.** The measures  $\delta_x \times P_{\Theta|X=x}$  are conditional distributions of  $P$  given  $X = x$  where  $X : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$  is the projection. The conditional expectation of an integrable  $\psi : \mathcal{X} \times \Theta \rightarrow \mathbb{R}$  given  $X = x$  is given by

$$E(\psi|X = x) = \frac{1}{g(x)} \int_{\Theta} \psi(x, \theta) f(x|\theta) d\mu_\Theta(\theta).$$

*Proof.* Let  $E'(\psi|X = x)$  denote the right hand side. We first show that it is well defined for almost all  $x \in \mathcal{X}$ . This follows since by assumption we have that  $|\psi(x, \theta)|$  is integrable with respect to  $P$  and thus

$$\int |\psi(x, \theta)| f(x|\theta) d\nu(x) d\mu_\Theta(\theta) < \infty.$$

It follows from Fubini-Tonelli that for  $\nu$  almost all  $x \in \mathcal{X}$  we have that

$$\int_{\mathcal{X}} |\psi(x, \theta)| f(x|\theta) d\mu_\Theta(\theta) < \infty$$

and thus  $E'(\psi|X = x)$  is a well defined measurable function on  $X$ . We have to show that  $(x, \theta) \mapsto E'(\psi|X = x)$  is measurable with respect to the  $\sigma$ -algebra induced by  $X : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$  and that for all  $A \subset X$  we have that

$$\int_{A \times \Theta} E'(\psi|X = x) dP(x, \theta) = \int_{A \times \Theta} \psi(x, \theta) dP(x, \theta).$$

The measurability is obvious as there is no dependence on  $\Theta$ . For the second claim, note that since (as shown above)  $g = dP_X/d\nu$ , we have

$$\begin{aligned} \int_{A \times \Theta} E'(\psi|X = x) dP(x, \theta) &= \int_A E'(\psi|X = x) dP_X(x) \\ &= \int_A E'(\psi|X = x) g(x) d\nu(x) \\ &= \int_A \int_{\Theta} \psi(x, \theta) f(x|\theta) d\mu_{\Theta}(\theta) d\nu(x) \end{aligned}$$

which completes the proof of this property since  $f(x|\theta) = \frac{dP}{d(\nu \times \mu_{\Theta})}(x, \theta)$ . Finally, observe that

$$\int_{\mathcal{X} \times \Theta} \psi d(\delta_x \times P_{\Theta|X=x}) = \int_{\Theta} \psi(x, \theta) dP_{\Theta|X=x}(\theta) = \frac{1}{g(x)} \int_{\Theta} f(x|\theta) \psi(x, \theta) d\mu_{\Theta}(\theta) = E(\psi | X = x)$$

so indeed  $\delta_x \times P_{\Theta|X=x}$  is the conditional probability of  $P$  given  $X = x$ .  $\square$

**Proposition 2.4.** For  $A \subset \mathcal{X}$  we have that

$$P(A \times \Theta | \Theta = \theta) = P_{\theta}(A).$$

*Proof.* By symmetry, if we swap the roles of  $\mathcal{X}$  with  $\Theta$  and apply the proposition above, then we get

$$\begin{aligned} P(A \times \Theta | \Theta = \theta) &= \left( \int_{\mathcal{X}} f(x|\theta) d\nu(x) \right)^{-1} \left( \int_A f(x|\theta) d\nu(x) \right) \\ &= P_{\theta}(X)^{-1} P_{\theta}(A) \\ &= P_{\theta}(A). \end{aligned}$$

$\square$

**2.1. Notation.** We use the standard notation from probability theory: if  $F_i : \Omega \rightarrow \Omega_i$  are measurable and  $P$  is a measure on  $\Omega$ , then

$$P(F_1 \in A_1, \dots, F_n \in A_n) := P(\{\omega \in \Omega \mid F_1(\omega) \in A_1, \dots, F_n(\omega) \in A_n\})$$

and likewise for conditional measures.

Thus if  $P$  is the measure on  $\mathcal{X} \times \Omega$  as constructed above, then

$$P(X \in A, \Theta \in B) = P(A \times B).$$

### 3. SUFFICIENT STATISTICS

**3.1. Basic definitions and examples.** A statistic is a measurable function  $T : \mathcal{X} \rightarrow \mathcal{T}$  to some measurable space  $(\mathcal{T}, \mathcal{B}_{\mathcal{T}})$ .

**Definition 3.1.** We say that a statistic  $T$  on  $\mathcal{X}$  is *sufficient (in the Bayesian sense)* if for any prior  $\mu_{\Theta}$  on  $\Theta$  we have for any Borel  $B \subset \Theta$  we have the equality

$$P(\Theta \in B \mid X = x) = P(\Theta \in B \mid T \circ X = Tx) \quad \text{for } \mu_X\text{-almost all } x \in \mathcal{X}.$$

**Example 3.2.** Consider the case where  $\mathcal{X} = \{0, 1\}^n$  and  $\Theta = [0, 1]$ . Suppose that  $P_\theta$  is the Binomial probability distribution on  $\{0, 1\}^n$  where  $(X_1, \dots, X_N) \in \mathcal{X}$  is drawn i.i.d with  $P(X_i = 1) = \theta$ . Let us for now assume the Prior distribution  $\mu_\Theta$  is unknown. We know that

$$P_\theta(X = x) = \theta^{T(x)}(1 - \theta)^{n-T(x)}$$

where  $T(x) = \sum_{i=1}^n x_i$  is the number of successes. The posterior distribution is thus given by

$$P(\Theta \in B \mid X = x) = \frac{P(\Theta \in B, X = x)}{P(X = x)} = \frac{\int_B \theta^{T(x)}(1 - \theta)^{n-T(x)} d\mu_\Theta(\theta)}{\int_\Theta \theta^{T(x)}(1 - \theta)^{n-T(x)} d\mu_\Theta(\theta)}$$

for all  $x \in \mathcal{X}$  that occur with positive probability (this is all of  $\mathcal{X}$  except for possibly  $x = (0, \dots, 0)$  or  $x = (1, \dots, 1)$  in case  $\theta$  is concentrated on 1 or 0 respectively).

We claim that  $T : \mathcal{X} \rightarrow \mathbb{Z}_{\geq 0}$  given by  $T(x) = \sum_{i=1}^n x_i$  is a sufficient statistic. Indeed if we fix  $x_0 \in \mathcal{X} = \{0, 1\}^n$  and we let  $m = Tx_0$ , we get that

$$\begin{aligned} P(\Theta \in B \mid T \circ X = m) &= \frac{P(\Theta \in B, T \circ X = m)}{P(T \circ X = m)} \\ &= \frac{\sum_{Tx=m} P(\Theta \in B, X = x)}{\sum_{Tx=m} P(X = x)} \\ &= \frac{\sum_{Tx=m} \int_B \theta^{T(x)}(1 - \theta)^{n-T(x)} d\mu_\Theta(\theta)}{\sum_{Tx=m} \int_\Theta \theta^{T(x)}(1 - \theta)^{n-T(x)} d\mu_\Theta(\theta)} \\ &= \frac{\binom{n}{m} \int_B \theta^m (1 - \theta)^{n-m} d\mu_\Theta(\theta)}{\binom{n}{m} \int_\Theta \theta^m (1 - \theta)^{n-m} d\mu_\Theta(\theta)} \end{aligned}$$

And so we see that this fraction cancels out to the expression for  $P(\Theta \in B \mid X = x_0)$  that we already obtained.

In fact, the same argument shows that  $T$  is still a sufficient statistic if instead of assuming that  $P_\theta$  is binomial we assume that  $P_\theta$  is *exchangeable*, meaning a distribution on  $\mathcal{X} = \{0, 1\}^n$  that is invariant under permutations of the co-ordinates, in this case the same calculation will instead yield

$$P(\Theta \in B \mid TX = m) = \frac{\sum_{Tx=m} \int_B P_\theta(x) d\mu_\Theta}{\sum_{Tx=m} \int_\Theta P_\theta(x) d\mu_\Theta}$$

but since we assume that  $P_\theta(x) = P_\theta(x')$  whenever  $x$  and  $x'$  are permutations of each other, we must have that all the summands are the same and hence we get the same cancellation.

**Remark 3.3.** Intuitively, sufficiency of a statistic essentially means that if we want to estimate the Posterior distribution of  $\theta$  given an observable statistic, we only need to look at the statistic of  $T \circ X$  rather than  $X$ , which may be simpler. In other words, somebody who wishes to estimate  $\theta$  (e.g., find a confidence interval) need not know the sample  $x$  but just needs to know  $Tx$ . In this example, the person wishing to estimate  $\theta$  needs only to be told a single integer  $Tx$ , rather than a potentially very long sequence  $x \in \{0, 1\}^n$  (data compression).

The classical definition of sufficiency does not make reference to a prior distribution and is thus not Bayesian.

**Definition 3.4.** We say that a statistic  $T : \mathcal{X} \rightarrow \mathcal{T}$  is *sufficient in the classical sense* if we have a function  $r : \mathcal{B} \times \mathcal{T} \rightarrow [0, 1]$  such that for all  $\theta \in \Theta$  and  $B \in \mathcal{B}$  (recall  $\mathcal{B}$  is the Borel  $\sigma$ -algebra on  $\mathcal{X}$ ) we have that

$$P_\theta(B \mid T = t) = r(B, t) \quad \text{for } T^*P_\theta - \text{almost all } t \in \mathcal{T}$$

and  $t \mapsto r(B, t)$  is Borel on  $\mathcal{T}$ .

This is a technical way of saying that  $T$  is sufficient if and only if  $P_\theta(B|T = t)$  does not depend on  $\theta$ . We stated the definition formally in order to deal with the issue that conditional probability is only almost everywhere unique and well defined.

**Example 3.5.** Let us continue from the binomial distribution in Example 3.2. As  $\mathcal{X} = \{0, 1\}^n$  is a finite set, we only have to consider the case where  $B \subset \mathcal{X}$  is a singleton  $B = \{x_0\}$  for some  $x_0 \in \mathcal{X}$ . Now  $P_\theta(B|T = t) = 0$  if  $Tx_0 \neq t$ . If  $Tx_0 = t$  then we have that

$$P_\theta(B|T = t) = \frac{P_\theta\{x_0\}}{P_\theta\{x \in \mathcal{X} \mid Tx = x_0\}} = \frac{1}{\binom{n}{t}}$$

where the last equality follows from the fact that all  $x \in \mathcal{X}$  with  $Tx = t$  have the same probability in a binomial distribution. In fact, the same argument applies to the case where the  $P_\theta$  are all exchangeable.

Thus we may take

$$r(\{x_0\}, t) = \mathbb{1}(Tx_0 = t) \binom{n}{t}^{-1}$$

and extend  $r$  to  $\mathcal{B} \times \mathbb{Z}$  by additivity. Thus this  $T$  is indeed sufficient in the classical sense.

**3.2. Equivalence of definitions.** In this section we prove equivalence of the classical and Bayesian definition of a sufficient statistics under the nice assumptions of Definition 1.5 (which we always assume).

**Lemma 3.6.** Suppose  $T : \mathcal{X} \rightarrow \mathcal{T}$  is a statistic and  $\mu_\Theta$  is a prior on  $\Theta$ . Suppose also that there is a Borel map  $s : \mathcal{T} \times \Theta \rightarrow [0, 1]$  so that

$$s(t, \theta) = P_\theta(A \mid T = t) \quad \text{for } T^*P_\theta\text{-almost all } t \in \mathcal{T}.$$

We have that

$$P(X \in A \mid TX = Tx, \Theta = \theta) = s(Tx, \theta) \quad \text{for } P\text{-almost all } (x, \theta) \in \mathcal{X} \times \Theta.$$

*Proof.* We must show that for  $B \subset \Theta, C \subset \mathcal{T}$  we have that

$$\int_{T^{-1}C \times B} P(A \times \Theta \mid Tx = t, \Theta = \theta) dP(x, \theta) = \int_{T^{-1}C \times B} s(Tx, \theta) dP(x, \theta).$$

The law of total probability tells us that the left hand side is equal to

$$P((A \times \Theta) \cap (T^{-1}C \times B)) = P(X \in A \cap T^{-1}C, \Theta \in B).$$

The right hand side is

$$\begin{aligned} \int_B \left( \int_{T^{-1}C} P_\theta(A|T = Tx) dP_\theta(x) \right) d\mu_\Theta(\theta) &= \int_B P_\theta(A \cap T^{-1}C) d\mu_\Theta(\theta) \\ &= \int_B \int_{\mathcal{X}} \mathbb{1}_{A \cap T^{-1}C}(x) dP_\theta(x) d\mu_\Theta(\theta) \\ &= P(X \in A \cap T^{-1}C, \Theta \in B) \end{aligned}$$

□

**Theorem 3.7.** Suppose that  $T : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient in the classical sense. Then  $T$  is also sufficient in the Bayesian sense (for any prior  $\mu_\Theta$  on  $\Theta$ ). Moreover for  $A \subset \mathcal{X}$  we have

$$(1) \quad P(X \in A \mid TX = Tx, \Theta = \theta) = P(X \in A \mid TX = Tx) \quad \text{for } P\text{-almost all } (x, \theta) \in \mathcal{X} \times \Theta.$$

*Proof.* We first show the *Moreover* claim. By the definition of sufficiency we have that

$$P_\theta(A \mid T = t) = r(A, t)$$

for some measurable  $t \mapsto r(A, t)$  and so indeed the Borel condition of the previous Lemma is satisfied. By the previous Lemma the left hand side of (1) is equal to  $r(A, Tx)$ . So now it remains to show that  $r(A, Tx) = P(A \times \Theta \mid TX = Tx)$  for  $P$ -almost all  $(x, \theta)$ . Thus we must show that for  $C \subset \mathcal{T}$  we have that

$$(2) \quad \int_{T^{-1}C \times \Theta} r(A, Tx) dP(x, \theta) = \int_{T^{-1}C \times \Theta} P(A \times \Theta \mid TX = Tx) dP(x, \theta).$$

By definition of conditional probability w.r.t  $TX : \mathcal{X} \times \Theta \rightarrow \mathcal{T}$ , the right hand side is equal to

$$P(A \times \Theta \cap T^{-1}C \times \Theta) = P(X \in A \cap T^{-1}C).$$

Repeating the calculation of the previous Lemma we have that the left hand side of (2) is

$$\begin{aligned} \int_{\Theta} \left( \int_{T^{-1}C} P_\theta(A \mid T = Tx) dP_\theta(x) \right) d\mu_\Theta(\theta) &= \int_{\Theta} P_\theta(A \cap T^{-1}C) d\mu_\Theta(\theta) \\ &= \int_{\Theta} \int_{\mathcal{X}} \mathbb{1}_{A \cap T^{-1}C}(x) dP_\theta(x) d\mu_\Theta(\theta) \\ &= P(X \in A \cap T^{-1}C). \end{aligned}$$

This completes the proof of (2). Now it remains to show that  $T$  is sufficient in the Bayesian sense. By this just proved *Moreover* claim, it follows from Lemma D.4 that  $X$  and  $\Theta$  are conditionally independent given  $TX$ . Using Lemma D.4 again we have that

$$P(\Theta \in B \mid TX = Tx, X = x) = P(\Theta \in B \mid TX = Tx) \quad \text{for } P\text{-almost all } (x, \theta)$$

and thus for  $P_X$ -almost all  $x \in \mathcal{X}$ . But the left hand side is clearly  $P(\Theta \in B \mid X = x)$ . Thus  $T$  is indeed sufficient in the Bayesian sense.  $\square$

**Lemma 3.8.** A statistic  $T : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient in the Bayesian sense if and only if for all prior distributions  $\mu_\Theta$  on  $\Theta$  and measurable  $B \subset \mathcal{X}$  we have that

$$x \mapsto P(\Theta \in B \mid x \in X)$$

is almost surely  $T^{-1}\mathcal{B}_T$ -measurable (i.e., almost surely equal to a function of  $T$ ).

*Proof.* We prove the non-obvious direction only: Thus suppose that for each measurable  $B \subset \mathcal{X}$  we have that  $x \mapsto P(\Theta \in B \mid x \in X) = h_B(Tx)$  where  $h_B : \mathcal{T} \rightarrow \mathbb{R}$  is Borel. We first use apply the law of total probability to the map  $T \circ X : \mathcal{X} \times \Theta \rightarrow \mathcal{T}$  as follows. For  $A \subset \mathcal{T}$  Borel we have that

$$P(\Theta \in B, T \circ X \in A) = \int_A P(\Theta \in B \mid T \circ X = t) dP_T(t).$$

Now we apply the law of total probability to the map  $X : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$  to obtain that

$$\begin{aligned}
P(\Theta \in B, T \circ X \in A) &= P(\Theta \in B, X \in T^{-1}A) \\
&= \int_{T^{-1}A} P(\Theta \in B | X = x) dP_X(x) \\
&= \int_{T^{-1}A} h_B(Tx) dP_X(x) \\
&= \int_A h_B(t) dP_T(t).
\end{aligned}$$

Thus we must have that  $P(\Theta \in B | T \circ X = t) = h_B(t)$  holds for almost all  $t$ . Thus

$$P(\Theta \in B | T \circ X = Tx) = h_B(Tx) = P(\Theta \in B | X = x)$$

holds for almost all  $x \in X$ .  $\square$

**Lemma 3.9.** Suppose that  $\nu$  is some  $\sigma$ -finite measure on  $\mathcal{X}$  such that  $P_\theta \ll \nu$  for all  $\theta \in \Theta$ . If  $T : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient in the Bayesian sense then there exists a probability measure  $\nu^*$  on  $\mathcal{X}$  such that  $P_\theta \ll \nu^*$  for all  $\theta \in \Theta$ ,  $\nu^* \ll \nu$  and for each  $\theta \in \Theta$  there is a Borel

$$h_\theta : \mathcal{T} \rightarrow [0, \infty)$$

such that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx) \quad \text{for } \nu^*\text{-almost all } x.$$

*Proof.* By Lemma E.2 there exists a countable or finite sequence of distinct  $\theta_1, \theta_2, \dots$  and  $0 < c_1, c_2, \dots \leq 1$  with  $\sum_i c_i = 1$  such that  $\nu^* := c_i P_{\theta_i}$  satisfies that  $P_\theta \ll \nu^* \ll \nu$  for all  $\theta \in \Theta$ . Now fix any  $\theta_0 \in \Theta$ . First suppose  $\theta_0 \neq \theta_i$  for all  $i > 0$ . Define now a prior  $\mu$  on  $\Theta$  where  $\mu(\{\theta_0\}) = 1/2$  and  $\mu(\{\theta_i\}) = c_i/2$ , i.e.,  $\mu$  is discrete. Now we have

$$\begin{aligned}
P(\Theta = \theta_0 | X = x) &= \frac{\int_{\{\theta_0\}} f(x|\theta) d\mu(\theta)}{\int_{\Theta} f(x|\theta) d\mu(\theta)} \\
&= \frac{\frac{1}{2} f(x|\theta_0)}{\frac{1}{2} f(x|\theta_0) + \frac{1}{2} \sum_{i>0} c_i f(x|\theta_i)}.
\end{aligned}$$

Observe that the pushforward measure on  $\mathcal{X}$  is  $P_X = \frac{1}{2} P_{\theta_0} + \frac{1}{2} \nu^*$ . We know by Lemma 2.2 that for  $P_X$ -almost all  $x \in X$  the denominator is non-zero. Now observe that

$$\frac{d\nu^*}{d\nu}(x) = \sum_{i>0} c_i f(x|\theta_i)$$

and as  $\nu^* \ll \nu$  we have that this is  $\nu^*$  almost surely non-zero and thus  $P_X$ -almost surely non-zero as  $P_X \ll \nu^*$  as shown above. So we can divide by this quantity to get that

$$(3) \quad \frac{dP_{\theta_0}}{d\nu^*(x)} = f(x|\theta_0) \left( \sum_{i>0} c_i f(x|\theta_i) \right)^{-1} \quad \text{for } P_X\text{-almost all } x \in \mathcal{X}.$$

Thus we have that

$$P(\Theta = \theta_0 | X = x) = \frac{dP_{\theta_0}/d\nu^*(x)}{dP_{\theta_0}/d\nu^*(x) + 1}$$

holds  $P_X$ -almost surely. However,  $P_X$ -almost surely we have that the right hand side is  $\mathcal{T}$ -measurable. Thus  $\nu^*$ -almost surely it also is. This completes the proof in the case  $\theta_0 \neq \theta_i$  for all  $i > 0$ . If  $\theta_0 = \theta_i$  for some  $i > 0$  we instead define the prior

$$\mu = \sum_{i>0} c_i P_{\theta_i}.$$

Without loss of generality, say  $\theta_0 = \theta_1$ . Thus

$$P(\Theta = \theta_1 \mid X = x) = \frac{c_1 f(x|\theta_1)}{\sum_{i>0} c_i f(x|\theta_i)} = c_1 \frac{dP_{\theta_1}}{d\nu^*}(x).$$

Thus again we have that this is  $P_X$ -almost equal to a  $\mathcal{T}$ -measurable function (and thus  $\nu^*$  almost equal to such).

□

**Theorem 3.10.** Assume the nice condition of Definition 1.5 . If  $T : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient in the Bayesian sense, then it is also sufficient in the classical sense.

*Proof.* Let  $\nu^*$  be the measure from the previous Lemma and recall that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx).$$

We will now show that for all  $\theta \in \Theta$  and  $A \subset X$  we have that

$$P_\theta(A \mid T = t) = \nu^*(A \mid T = t),$$

which will complete the proof. Observe that for  $C \subset \mathcal{T}$  we have that

$$\begin{aligned} \int_C \nu^*(A \mid T = t) d(T^* P_\theta)(t) &= \int_{T^{-1}C} \nu^*(A \mid T = Tx) dP_\theta(x) \\ &= \int_{T^{-1}C} \nu^*(A \mid T = Tx) h_\theta(Tx) d\nu^*(x) \\ &= \int_{\mathcal{X}} \mathbb{E}(\mathbb{1}_A \mid T = Tx) \mathbb{1}_{T^{-1}C}(x) h_\theta(Tx) d\nu^*(x) \\ &= \int_{\mathcal{X}} \mathbb{1}_A(x) \mathbb{1}_{T^{-1}C}(x) h_\theta(Tx) d\nu^*(x) \\ &= \int_{\mathcal{X}} \mathbb{1}_A(x) \mathbb{1}_{T^{-1}C}(x) dP_\theta(x) = P_\theta(A \cap T^{-1}C) \end{aligned}$$

which completes the proof by the law of total probability.

□

By observing the proofs so far, we see that to check sufficiency in the Bayesian sense it is enough to check only those prior distributions that are discrete (i.e., those  $\mu_\Theta$  on  $\Theta$  where  $\mu(C) = 1$  for some at most countable subset  $C \subset \Theta$ ).

**Corollary 3.11.** Assuming the nice condition of Definition 1.5 . We have that  $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient (in either classical or Bayesian sense) if and only if the Bayesian sufficiency condition is satisfied for all discrete prior distributions  $\mu_\Theta$  on  $\Theta$ .

*Proof.* Observe that the measures  $\mu$  used in the proof of Lemma 3.9 are all discrete. Once  $\nu^*$  is constructed, sufficiency follows as seen in the proof of the previous Theorem.

□

**3.3. Fisher-Neyman factorization.** There is a convenient characterization of sufficiency.

**Theorem 3.12.** Assume the nice condition of Definition 1.5. Then a statistic  $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{T}$  is sufficient if and only if the following holds: There is Borel  $g : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$  such that for all  $\theta \in \Theta$  we have a Borel  $h_\theta : \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}$  such that

$$f(x|\theta) = h_\theta(\mathcal{T}x)g(x) \quad \text{for } P_\theta \text{ almost all } x \in X$$

*Proof.* First suppose this condition holds. Thus by Corollary 3.11 we only need to check Bayesian sufficiency for any discrete prior  $\mu$  on  $\Theta$ . Say  $\mu$  is supported on  $C = \{\theta_1, \theta_2, \dots\} \subset \Theta$ . For  $C' \subset C$  we have by Bayes' formula that

$$P(\Theta \in C' \mid X = x) = \frac{\int_{C'} f(x|\theta) d\mu(\theta)}{\int_\Theta f(x|\theta) d\mu(\theta)} = \frac{\sum_{\theta \in C'} h_\theta(\mathcal{T}x)g(x)\mu(\{\theta\})}{\sum_{\theta \in C} h_\theta(\mathcal{T}x)g(x)\mu(\{\theta\})} = \frac{\sum_{\theta \in C'} h_\theta(\mathcal{T}x)\mu(\{\theta\})}{\sum_{\theta \in C} h_\theta(\mathcal{T}x)\mu(\{\theta\})}.$$

Observe now that this is  $\mathcal{T}$ -measurable since the sums are countable, thus Bayesian sufficiency is established.

(**Remark:** this is the reason we restrict without loss of generality to  $\mu$  discrete, as otherwise it seems one cannot show measurability in the integrand unless we also assume that  $(\theta, t) \mapsto h_\theta(t)$  is Borel, which we do not want to as we have not been able to verify this in Lemma 3.9).

Conversely, suppose that  $\mathcal{T}$  is Bayesian. Thus by Lemma 3.9) we have

$$f(x|\theta) = \frac{dP_\theta}{d\nu}(x) = \frac{dP_\theta}{d\nu^*}(x) \frac{d\nu^*}{d\nu}(x) = h_\theta(\mathcal{T}x)g(x)$$

for  $\nu^*$ -almost all  $x \in X$ . As  $P_\theta \ll \nu^*$  we have it also holds for  $P_\theta$ -almost all  $x \in X$ .  $\square$

**Example 3.13.** In Example 3.2 we have that

$$f(x|\theta) = P_\theta(X = x) = \theta^{T(x)}(1 - \theta)^{n - T(x)}$$

so can just take

$$h_\theta(t) = \theta^t(1 - \theta)^{n - t}$$

and  $g(x) = 1$ . Here  $\nu(\{x\}) = 1$  is the counting measure. If instead we had a different measure  $\nu(\{x\}) = q(x) > 0$  then we would have

$$f(x|\theta) = P_\theta(X = x)q(x)^{-1} = h_\theta(\mathcal{T}x)q(x)^{-1}$$

so then  $g(x) = q(x)^{-1}$ . This shows that  $g(x)$  is just a change of density.

## APPENDIX A. REVIEW OF PROBABILITY

### APPENDIX B. BOREL SPACES

**Lemma B.1.** Let  $X$  be a second-countable metric space. Then there exists an embedding  $\iota : X \hookrightarrow Y$  where  $Y$  is compact and  $\iota$  is a homeomorphism onto its image  $\iota(X)$ .

*Proof.* Let  $u_1, u_2, \dots$  be a countable dense subset of  $X$ . Suppose that  $d(x, x') \leq 1$  for all  $x, x'$ . Define

$$\iota : X \rightarrow [0, 1]^{\mathbb{N}}$$

to be

$$\iota(x) = (d(x, u_i))_{i=1}^{\infty}$$

where  $Y = [0, 1]^{\mathbb{N}}$  has the product topology, thus is a compact metric space. Continuity and injectivity of  $\iota$  is clear. Now suppose that  $x_1, x_2, \dots \in X$  and  $x \in X$  are such that  $\iota(x_i) \rightarrow \iota(x)$  as  $i \rightarrow \infty$ . Fix  $\epsilon > 0$  and

choose  $u_j$  so that  $d(x, u_j) < \epsilon$ . Now for all sufficiently large  $i$  we have that  $|d(x_i, u_j) - d(x, u_j)| < \epsilon$ . This implies that  $d(x_i, u_j) < 2\epsilon$ . Now

$$d(x, x_i) \leq d(x, u_j) + d(u_j, x_i) \leq \epsilon + 2\epsilon = 3\epsilon.$$

Thus  $\iota^{-1}$  is continuous on  $\iota(X)$ .  $\square$

**Lemma B.2.** Let  $X$  be a complete metric space, with metric  $d_X$ , and suppose that  $Y \subset X$  is a subset of  $X$  such that the subspace topology on  $Y$  is induced by a complete metric  $d_Y$  on  $Y$ . Then  $Y$  is a countable intersection of open subsets of  $X$ , and is thus a Borel subset of  $Y$ .

*Proof.* We first show that for each  $\epsilon > 0$  and  $y \in Y$  there exists an open set  $V_{y,\epsilon}$  of  $X$  such that  $y \in V$  and  $\text{diam}_{d_X}(V_{y,\epsilon}) \leq \epsilon$  and  $\text{diam}_{d_Y}(V_{y,\epsilon} \cap Y) \leq \epsilon$ . To see this, observe since  $d_X$  and  $d_Y$  induce the same topology on  $Y$ , there is an open set  $\tilde{V}$  of  $X$  such that  $\tilde{V} \cap Y$  is the ball of radius  $\epsilon/2$  around  $y$  in the  $d_Y$  metric. Hence we can take

$$V_{y,\epsilon} = \tilde{V} \cap B_{d_X}(y, \epsilon/2).$$

Now let  $U_n = \bigcup_{y \in Y} V_{y,1/n}$  (Remark: to avoid axiom of choice, just take the union of  $y$  and all the different sets of the form  $\tilde{V}$ ). Observe now that  $Y \subset U_n$  for all  $n$ . We claim that

$$Y = \bigcap_{n=1}^{\infty} U_n.$$

Thus suppose that  $x \in U_n$  for all  $n \geq 1$ . We must show  $x \in Y$ . We will show that, by induction on  $n$ , that there exists a sequence of open subsets  $W_1 \supset W_2 \supset \dots$  of  $X$  and  $y_n \in Y \cap W_n$  such that  $\text{diam}_{d_X}(W_n) \leq \frac{1}{n}$ ,  $\text{diam}_{d_Y}(W_n \cap Y) \leq \frac{1}{n}$ ,  $x \in W_n$ . We can take  $W_1 = V_{y_1,1}$  for some  $y_1 \in Y$  since  $x \in U_1$ . Now suppose  $W_n$  has been constructed. Thus there exists  $\epsilon > 0$  so that  $B_{d_X}(x, \epsilon) \subset W_n$  as  $W_n$  is open and  $x \in W_n$ . Now take  $N > n$  such that  $1/N < \epsilon$ . Now since  $x \in U_N$  there exists  $y_{n+1} \in Y$  and  $W_{n+1} \subset X$  open so that  $y_{n+1}, x \in W_{n+1}$  and  $\text{diam}_{d_X}(W_{n+1}) \leq 1/N$ ,  $\text{diam}_{d_Y}(W_{n+1} \cap Y) \leq 1/N$ . Now observe that  $W_{n+1} \subset B_{d_X}(x, \epsilon) \subset W_n$  since if  $x' \in W_{n+1}$  then

$$d_X(x, x') \leq \text{diam}_{d_X}(W_{n+1}) \leq 1/N < \epsilon.$$

Thus in summary  $y_{n+1} \in W_{n+1}$  is non-empty as desired and  $W_{n+1} \subset W_n$  and

$$\text{diam}_{d_Y}(W_{n+1} \cap Y) \leq \frac{1}{N} \leq \frac{1}{n+1}.$$

It now follows that  $y_n \rightarrow x$  in  $d_X$  metric and that  $y_1, y_2, \dots$  is a Cauchy sequence in  $Y$ . Thus by completeness, we have that  $y_n \rightarrow y \in Y$  for some  $y \in Y$ . However

$$d_X(x, y) \leq d_X(x, y_n) + d_X(y_n, y) \rightarrow 0$$

as  $n \rightarrow \infty$  since  $d_Y(y, y_n) \rightarrow 0$  and  $d_X$  and  $d_Y$  induce the same topology. Thus  $x = y \in Y$  as required.  $\square$

**Corollary B.3.** Let  $X$  be a second countable complete metric space. Then  $X$  is homeomorphic to a Borel subset of a compact metric space. In fact, we can take this Borel subset to be a countable intersection of open subsets.

**Definition B.4.** A *standard Borel space* is a measurable space  $(X, \mathcal{B})$  where  $X$  is a second-countable complete metric space and  $\mathcal{B}$  is the Borel  $\sigma$ -algebra.

## APPENDIX C. CONDITIONAL EXPECTATION AND PROBABILITY

## C.1. Conditional expectation.

**Theorem C.1.** Let  $(X, \mathcal{B}, \mu)$  be a probability space and let  $\mathcal{A} \subset \mathcal{B}$  be a sub- $\sigma$ -algebra of  $\mathcal{B}$ . Then for  $f \in L^1(X, \mathcal{B}, \mu)$  there exists a unique  $g \in L^1(X, \mathcal{A}, \mu)$  such that

$$\int_A f d\mu = \int_A g d\mu \quad \text{for all } A \in \mathcal{A}.$$

**Definition C.2.** We say that the  $g \in L^1(X, \mathcal{B}, \mu)$  the *conditional expectation of  $f$  w.r.t.  $\mathcal{A}$* , and denote it by  $g = E(f|\mathcal{A})$ .

*Proof.* If  $f \in L^2(X, \mathcal{B}, \mu)$  then we define  $E(f|\mathcal{A})$  to be the orthogonal projection of  $f$  onto the closed subspace  $L^2(X, \mathcal{A}, \mu)$ . Indeed, for  $A \in \mathcal{A}$  we have that  $\mathbf{1}_A \in L^2(X, \mathcal{A}, \mu)$  and so

$$\int_A f d\mu = \langle f, \mathbf{1}_A \rangle = \langle E(f|\mathcal{A}), \mathbf{1}_A \rangle = \int_A E(f|\mathcal{A}) d\mu$$

as required. We will now extend  $E(\cdot|\mathcal{A})$  to  $L^1$ . By density of  $L^2$  in  $L^1$  it is enough to show that  $E(\cdot|\mathcal{A})$  is a bounded operator w.r.t. to the  $L^1$  norm on  $L^2$ . For  $f \in L^2(X, \mathcal{B}, \mu)$  let  $\psi \in L^\infty(X, \mathcal{A}, \mu)$  be such that  $|E(f|\mathcal{A})| = \psi E(f|\mathcal{A})$  and  $\|\psi\|_\infty \leq 1$ . Thus

$$\|E(f|\mathcal{A})\|_1 = \int_X \psi E(f|\mathcal{A}) d\mu = \langle E(f|\mathcal{A}), \overline{\psi} \rangle = \langle f, \overline{\psi} \rangle = \int_X \psi f d\mu \leq \|f\|_1.$$

Thus  $E(\cdot|\mathcal{A})$  has  $L^1$  operator norm at most 1 and thus can be extended to  $L^1(X, \mathcal{B}, \mu) \rightarrow L^1(X, \mathcal{A}, \mu)$ .

For uniqueness, suppose that  $g_1, g_2 \in L^1(X, \mathcal{A}, \mu)$  are real valued functions such that

$$\int_A g_1 d\mu = \int_A g_2 d\mu \quad \text{for all } A \in \mathcal{A}.$$

Let

$$A_\epsilon = \{x \in X \mid g_1(x) \leq g_2(x) - \epsilon\}$$

and observe that  $A_\epsilon \in \mathcal{A}$ . Thus

$$\int_{A_\epsilon} g_2 = \int_{A_\epsilon} g_1 d\mu \leq \int_{A_\epsilon} g_2 d\mu - \epsilon \mu(A_\epsilon)$$

which implies that  $\mu(A_\epsilon) = 0$ . Thus  $g_1 \geq g_2$  almost everywhere. By symmetry,  $g_1 = g_2$  almost everywhere. For  $g_1, g_2$  complex valued uniqueness also follows by equating real and imaginary parts.  $\square$

**Definition C.3.** Let  $T : X \rightarrow Y$  be a measurable map between measure spaces  $(X, \mathcal{B}_X)$  and  $(Y, \mathcal{B}_Y)$ . We define  $T^{-1}\mathcal{B}_Y = \{T^{-1}B \mid B \in \mathcal{B}_Y\}$ .

**Proposition C.4.** Suppose that  $(X, \mathcal{B}_X, \mu_X)$  is a probability space,  $(Y, \mathcal{B}_Y)$  is a measurable space  $T : X \rightarrow Y$  is measurable. Then any function  $f : X \rightarrow \mathbb{R}$  that is  $T^{-1}\mathcal{B}_Y$  measurable (that is,  $f^{-1}B \in T^{-1}\mathcal{B}_Y$  for all Borel  $B \subset \mathbb{R}$ ) can be written as  $f = g \circ T$  for some measurable  $g : Y \rightarrow \mathbb{R}$ .

In other words, if we let  $\mu_Y = T^*\mu_X$  be the push-forward measure then we have an isometric embedding

$$L^p(Y, \mathcal{B}_Y, \mu_Y) \hookrightarrow L^p(X, \mathcal{B}_X, \mu_X)$$

given by  $g \mapsto g \circ T$  whose image is exactly  $L^p(\mathcal{X}, T^{-1}\mathcal{B}_Y, \mu_X)$ .

*Proof.* If  $f = \mathbb{1}_A$  for some  $A \in T^{-1}\mathcal{B}_Y$  then  $A = T^{-1}B$  for some  $B \in \mathcal{B}_Y$  and so we can take  $g = \mathbb{1}_B$ . For  $f \geq 0$  note that we can write

$$f = \sum_{i=1}^{\infty} c_i \mathbb{1}_{A_i}$$

for some  $c_i \geq 0$  and  $A_i \in T^{-1}\mathcal{B}_Y$  where the limit is pointwise and so  $\mathbb{1}_{A_i} = \mathbb{1}_{B_i} \circ T$  for some  $B_i \in \mathcal{B}_Y$ . Thus  $f = g \circ T$  where

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

converges pointwise as it is monotonic  $c_i > 0$ . For arbitrary measurable  $f$ , note that we can write it as a linear combination of non-negative functions.

Now note that for  $g \geq 0$  we have that

$$\int_X g \circ T d\mu_X = \int_Y g d\mu_Y$$

which can be shown as follows. Writing

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

and using the monotone convergence theorem we have that

$$\int_X g(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \int_X \mathbb{1}_{B_i}(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \mu_X(T^{-1}B_i) = \sum_{i=1}^{\infty} c_i \mu_Y(B_i) = \int_Y g(y) d\mu_Y(y).$$

This shows that for  $1 \leq p < \infty$  we have that

$$\|g \circ T\|_p^p = \int_X |g|^p(Tx) d\mu_X(x) = \int_Y |g|^p(y) d\mu_Y(y) = \|g\|_p^p$$

and so indeed the embedding is an isometry.  $\square$

This proposition motivates us to define conditional expectation with respect to a random variable (or factor map).

**Definition C.5.** Let  $(X, \mathcal{B}, \mu)$  be a probability space and suppose that  $T : X \rightarrow Y$  is measurable where  $(Y, \mathcal{B}_Y)$  is a measure space (i.e., a random variable). Then for  $f \in L^p(\mathcal{X}, \mathcal{B}, \mu)$  (for  $p = 1, 2, \infty$ ) we define a function on  $Y \rightarrow \mathbb{C}$  denoted  $y \mapsto E(f|T = y)$  that is characterized for  $\mu$ -almost all  $x$  by

$$E(f|\mathcal{T}^{-1}\mathcal{B}_Y)(x) = E(f|T = y) \quad \text{where } y = Tx.$$

In other words, the function  $E(f|\mathcal{T}^{-1}\mathcal{B}_Y) \in L^p(\mathcal{X}, \mathcal{T}^{-1}\mathcal{B}_Y, \mu)$  is identified with a function in  $L^p(Y, \mathcal{B}_Y, T^*\mu)$  using the isometry in the proposition above and this function at  $y \in Y$  is denoted  $E(f|T = y)$ . Note that  $E(f|T = y)$  is uniquely defined for  $T^*\mu_X$ -almost all  $y \in Y$ .

Note that  $E(f|T = y)$  can be characterized by the property that

$$\int_A E(f|T = y) d\mu_Y(y) = \int_{T^{-1}A} f(x) d\mu_X(x) \quad \text{for all } A \in \mathcal{B}_Y.$$

## C.2. Conditional Probability.

**Theorem C.6** (Conditional probability). Let  $X$  be a second-countable complete metric space with Borel  $\sigma$ -algebra  $\mathcal{B}$  and suppose  $\mathcal{A} \subset \mathcal{B}$  is a sub  $\sigma$ -algebra. Let  $\mu$  be a probability measure on  $(X, \mathcal{B})$ . Then there exists  $X' \in \mathcal{A}$  with  $\mu(X') = 1$  and a family of probability measures  $\mu_x^{\mathcal{A}}$  for  $x \in X'$  such that the following holds.

- For  $\mathcal{B}$ -measurable functions  $f : X \rightarrow \mathbb{C}$  such that  $\|f\|_{\infty} < \infty$ , we have that the mapping  $X' \rightarrow \mathbb{C}$  given by

$$x \mapsto \int_X f(u) d\mu_x^{\mathcal{A}}(u)$$

is  $\mathcal{A}$ -measurable. The same is true if we replace  $\|f\|_{\infty} < \infty$  with  $f(x) \in \mathbb{R}_{\geq 0}$  for all  $x \in X$ .

- For  $f : X \rightarrow \mathbb{C}$  with  $\|f\|_1 < \infty$  we have that<sup>1</sup>

$$E(f|\mathcal{A})(x) = \int_X f(u) d\mu_x^{\mathcal{A}}(u) \quad \text{for almost all } x \in X.$$

**Definition C.7.** We call such a family of measures  $\mu_x^{\mathcal{A}}$  *conditional probabilities of  $\mu$  w.r.t  $\mathcal{A}$* .

*Proof.* Using Corollary B.3 we may assume  $X$  is compact. Thus there exists a countable dense  $\mathbb{Q}$ -vector subspace  $V$  of  $C(X)$  such that  $1 \in V$ . Now for each  $f \in V$  choose a genuine function (not an equivalence class of functions)  $g_f : X \rightarrow \mathbb{C}$  such that

$$g_f(x) = E(f|\mathcal{A})(x)$$

and

$$\|g_f\|_{\infty} \leq \|f\|_{\infty}.$$

Now for any  $\alpha, \beta \in \mathbb{Q}$  and  $f_1, f_2 \in V$  we have that

$$g_{\alpha f_1 + \beta f_2}(x) = \alpha g_{f_1}(x) + \beta g_{f_2}(x) \quad \text{for almost all } x \in X.$$

We also have  $g_1(x) = 1$  and  $g_f(x) \geq 0$  for positive  $f \in V$  (i.e,  $f \geq 0$ ) for almost all  $x \in X$ . Thus we may choose  $X' \subset X$  to be a set with  $\mu(X') = 1$  on which all of these properties hold (as there are only countably many equations). Thus for each  $x \in X'$  we have a linear function  $\Lambda'_x : V \rightarrow \mathbb{C}$  given by

$$\Lambda'_x(f) = g_f(x).$$

Note that  $\Lambda'_x$  is uniformly continuous (since  $\|g_f\|_{\infty} \leq \|f\|_{\infty}$ ) and thus it extends to a linear functional  $\Lambda_x : C(X) = \overline{V} \rightarrow \mathbb{C}$  which also satisfies that

$$|\Lambda_x(f)| \leq \|f\|_{\infty}.$$

We wish to apply the Riesz representation theorem, thus it remains to show that  $\Lambda_x$  is a positive linear functional. Now suppose  $f \in C(X)$  is positive. Then there exists  $f_i \in V$  such that  $f_i \rightarrow f$  uniformly. Now fix a positive integer  $q$  and let  $\epsilon = 1/q$ . Then  $\Lambda_x(\epsilon) = \epsilon$  since  $g_1 = 1$ . For  $i$  sufficiently large we have that we have  $\|f_i - f\|_{\infty} < \epsilon$  and so  $f_i + \epsilon \geq 0$  as  $f$  is positive. As  $f_i + \epsilon \in V$  we have that

$$\Lambda_x(f_i) + \epsilon = g_{f_i + \epsilon}(x) \geq 0.$$

Thus

$$\Lambda_x(f) \geq \Lambda_x(f_i) - \Lambda_x(|f_i - f|) \geq \Lambda_x(f_i) - \epsilon > -2\epsilon.$$

---

<sup>1</sup>The right hand side may be undefined on a null set of  $x$ .

This shows that  $\Lambda_x(f) \geq 0$  and thus  $\Lambda_x$  is a positive linear continuous functional on  $C(X)$  with  $\Lambda_x(1) = 1$ . Thus there exists a measure  $\mu_x$  such that

$$\Lambda_x(f) = \int_X f(u) d\mu_x(u) \quad \text{for all } f \in C(X), x \in X'.$$

In particular this means that

$$E(f|\mathcal{A})(x) = \int_X f(u) d\mu_x(u) \quad \text{for all } f \in V.$$

This means that

$$(4) \quad x \mapsto \int_{X'} f(u) d\mu_x(u) \text{ is } \mathcal{A}\text{-measurable on } X'$$

and

$$(5) \quad \int_A \int_{X'} f(u) d\mu_x(u) d\mu(x) = \int_A f(x) d\mu(x) \quad \text{for all } A \in \mathcal{A}$$

holds for all  $f \in V$ . By density of  $V$  in  $C(X)$ , this also holds for all  $f \in C(X)$  and the integrand is indeed  $\mathcal{A}$  measurable (it is a pointwise limit of  $\mathcal{A}$  measurable functions). Since the indicator function of an open set is a monotonic limit of continuous functions, we have that (4) and (5) holds for  $f = \mathbb{1}_U$  for all  $U \subset X$  open. It also holds for closed sets by linearity. Finally, observe that it also holds for an intersection  $U \cap C$  where  $U$  is open and  $C$  is closed since such a set is a countable nested intersection of open sets.

Now let

$$\mathcal{R} = \{\cup_{i=1}^n U_i \cap C_i \mid U_1, \dots, U_n \text{ are open and } C_1, \dots, C_n \text{ are closed.}\}$$

be the set of all finite unions of sets that can be written as an intersection of an open set and a closed set. We claim that any element in  $\mathcal{R}$  can be written as a finite disjoint union of an intersection of an open set with a closed set. Indeed, any such finite union belongs to a finite  $\sigma$ -algebra generated by such sets but any atom of this  $\sigma$ -algebra must take this form. It now follows that (4) and (5) holds for  $f = \mathbb{1}_R$  for all  $R \in \mathcal{R}$ . By the monotone convergence theorem, these also hold on the monotone class generated by  $\mathcal{R}$ . Thus by the monotone class lemma, it holds on all Borel sets since  $\mathcal{R}$  is an algebra. Any measurable  $f : X \rightarrow \mathbb{R}_{\geq 0}$  is a monotonic limit of simple functions and so it holds for such  $f$  too, which completes the proof for such  $f$ . The theorem now follows for general  $f : X \rightarrow \mathbb{C}$  with  $\|f\|_1$  as desired by linearity.  $\square$

We now show that the conditional probabilities are almost surely unique and provide some characterizations that are practical to use. In the following proposition, we fix  $X, \mathcal{A}, \mu$  as in Theorem C.6.

**Proposition C.8.** If  $\mu_x^{\mathcal{A}}$  and  $\nu_x^{\mathcal{A}}$  are two conditional probabilities, for some  $x \in X' \in \mathcal{A}$  of full measure, then  $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$  for  $\mu$ -almost all  $x \in X$ .

*Proof.* We for each  $f \in C(X)$  we have that

$$\int_X f(u) d\mu_x^{\mathcal{A}}(u) = E(f|\mathcal{A})(x) = \int_X f(u) d\nu_x^{\mathcal{A}}(u)$$

holds for all  $x \in X_f \in \mathcal{A}$  with  $\mu(X_f) = 1$ . Applying this to a countable dense set of  $f \in C(X)$  we can find a single set  $X' \in \mathcal{A}$  with  $\mu(X') = 1$  so that this identity holds for all continuous functions  $f \in C(X)$ . It follows by uniqueness in Riesz-representation theorem that  $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$  for all  $x \in X'$ .  $\square$

**C.3. Conditional probability with respect to a random variable.** In this subsection,  $\mathcal{A}$  is a countable generated  $\sigma$ -algebra of  $\mathcal{B}$  where  $(X, \mathcal{B})$  is standard Borel. So  $\mathcal{A} = \sigma(\{A_1, A_2, \dots\})$  where  $A_1, A_2, \dots \in \mathcal{A}$ . For each  $x \in X$  we can define the atom

$$[x]_{\mathcal{A}} = \bigcap_{i, x \in A_i} A_i \cap \bigcap_{j, x \notin A_j} (X \setminus A_j)$$

containing  $x$ .

**Proposition C.9.** For  $x \in X$  we have that  $[x]$  is the smallest subset of  $\mathcal{A}$  containing  $x$ .

*Proof.* We must show that for each  $A \in \mathcal{A}$  we have either  $[x] \subset A$  or  $[x] \cap A = \emptyset$ . This is true for each  $A_i$  and so one must show that the sets in  $\mathcal{A}$  satisfying this property forms a  $\sigma$ -algebra.  $\square$

**Proposition C.10.** There is a set  $X'' \subset X$  with  $X'' \in \mathcal{A}$  and  $\mu(X'') = 1$  such that for any  $x_1, x_2 \in X''$  we have that  $\mu_{x_1}^{\mathcal{A}} = \mu_{x_2}^{\mathcal{A}}$  if and only if  $[x_1] = [x_2]$ . Moreover,  $\mu_x^{\mathcal{A}}([x]) = 1$  for  $x \in X''$ .

*Proof.* Let  $\mu_x = \mu_x^{\mathcal{A}}$  for  $x \in X'$  be a set of conditional probabilities, where  $X' \in \mathcal{A}$  and  $\mu(X') = 1$ . For almost all  $x \in X'$  we have that

$$\mu_x(A_i) = \int \mathbb{1}_{A_i} d\mu_x = E(\mathbb{1}_{A_i} | \mathcal{A})(x) = \mathbb{1}_{A_i}(x).$$

Say this occurs on  $X'' \subset X'$  with  $\mu(X'') = 1$ . Note that since all terms are  $\mathcal{A}$ -measurable, we have  $X'' \in \mathcal{A}$ . Now let  $x_1, x_2 \in X''$ . If  $\mu_{x_1} = \mu_{x_2}$  then  $\mathbb{1}_{A_i}(x_1) = \mathbb{1}_{A_i}(x_2)$  for all  $i$  and thus  $[x_1] = [x_2]$ . Conversely, suppose that  $[x_1] = [x_2]$ . Thus any  $\mathcal{A}$ -measurable function must have equal values on  $x_1$  and  $x_2$ . But for any  $f \in C(X)$  we have that

$$x \mapsto \int f d\mu_x$$

is  $\mathcal{A}$ -measurable. Thus

$$\int f d\mu_{x_1} = \int f d\mu_{x_2}$$

holds for all  $f \in C(X)$  and thus  $\mu_{x_1} = \mu_{x_2}$ .  $\square$

**Theorem C.11** (Law of total probability). Suppose that  $T : X \rightarrow Y$  is a measurable map between standard Borel spaces and let  $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$  be the induced  $\sigma$ -algebra. There exists Borel  $Y' \subset Y$  and probability measures  $\mu_y$  on  $X$ , for each  $y \in Y'$ , such that the following hold.

- $\mu(T^{-1}Y') = 1$
- For all  $y \in Y'$ , the measure  $\mu_y$  is supported on  $T^{-1}(\{y\})$ .
- For all  $x \in T^{-1}(Y')$  we have that  $\mu_{Tx} = \mu_x^{\mathcal{B}_T}$ . That is the family  $\mu_{Tx}$  for  $x \in T^{-1}(Y')$  is a family of conditional probability measures for the  $\sigma$ -algebra  $\mathcal{B}_T$ .

Moreover, for bounded Borel  $f : X \rightarrow \mathbb{C}$  and  $B \subset Y$  Borel we have that

$$\int_{T^{-1}B} f(x) d\mu(x) = \int_B \left( \int_X f(u) d\mu_y(u) \right) d(T^*\mu)(y).$$

*Proof.* The induced  $\sigma$ -algebra  $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$  is countably generated and its atoms are precisely the fibers

$$T^{-1}(\{y\}) \quad \text{for } y \in Y.$$

In other words,

$$[x]_{\mathcal{B}_T} = T^{-1}(\{Tx\}) \quad \text{for } x \in X.$$

We thus have by the above proposition that the conditional measure  $\mu_x^{\mathcal{B}_T}$  is supported on the fiber  $T^{-1}(\{Tx\})$  and only depends on the fiber. More precisely, we have a family of measures  $\mu_y$  on  $X$ ,  $y \in Y'$  for some Borel  $Y' \subset Y$  such that  $\mu_y$  is supported on  $T^{-1}(\{y\})$  and such that for bounded Borel  $f : X \rightarrow \mathbb{C}$  we have that

$$x \mapsto \int_X f(u) d\mu_{Tx}(u)$$

is  $\mathcal{B}_T$  measurable and

$$\int_A f(x) d\mu(x) = \int_A \int f(u) d\mu_{Tx}(u) d\mu(x) \quad \text{for } A \in \mathcal{B}_T.$$

Now by Proposition C.4 we have that there exists  $g : Y' \rightarrow \mathbb{C}$  which is Borel so that

$$g(Tx) = \int_X f(u) d\mu_{Tx}(u).$$

In particular, this means that

$$y \mapsto g(y) = \int_X f(u) d\mu_y(u)$$

is Borel on  $Y'$ .

Now for Borel  $B \subset Y'$  we have that  $A = T^{-1}B \in \mathcal{B}_T$  and the equation above says that

$$\begin{aligned} \int_A f(x) d\mu(x) &= \int_A g(Tx) d\mu(x) \\ &= \int_B g(y) d(T^*\mu)(y) \\ &= \int_B \left( \int_X f(u) d\mu_y(u) \right) d(T^*\mu)(y) \end{aligned}$$

which completes the proof. □

It is customary to define

$$\mu(B \mid T = y) := \mu_y(B)$$

and

$$E(f|T = y) := E(f|\mathcal{B}_T)(x)$$

for  $y = Tx$ . The law of total probability then says

$$\mu(A \cap T^{-1}B) = \int_B \mu(A|T = y) d\mu_Y(y)$$

for any Borel  $A \subset X, B \subset Y$  where  $\mu_Y = T^*\mu$ .

#### APPENDIX D. CONDITIONAL INDEPENDENCE

In this section, fix a standard Borel probability space  $(\Omega, \mu)$  and suppose we have maps  $X : \Omega \rightarrow \mathcal{X}$ ,  $Y : \Omega \rightarrow \mathcal{Y}$  and  $T : \Omega \rightarrow \mathcal{T}$  to some standard Borel spaces. We also let  $\mathcal{A}_X, \mathcal{A}_Y, \mathcal{A}_T$  be the  $\sigma$ -algebras induced by these maps, respectively.

**Definition D.1.** We say that  $X$  and  $Y$  are conditionally independent given  $T$  if for all  $A \in \mathcal{A}_X, B \in \mathcal{A}_Y$  we have that

$$\mu(X^{-1}A \cap Y^{-1}B | T = t) = \mu(X^{-1}A | T = t) \mu(Y^{-1}B | T = t) \quad \text{for } T^*\mu\text{-almost all } t \in \mathcal{T}.$$

In particular, if  $T$  is constant then we say that  $X$  and  $Y$  are independent, i.e.,  $X$  and  $Y$  are independent if and only if for all  $A \subset \mathcal{X}, B \subset \mathcal{Y}$  we have

$$\mu(X^{-1}A \cap Y^{-1}B) = \mu(X^{-1}A)\mu(Y^{-1}B).$$

**Example D.2.** Suppose  $\Omega = [0, 1] \times [0, 1] \cup [1, 2] \times [1, 2]$  is the disjoint union of two squares and  $\mu$  is the uniform probability measure on  $\Omega$ . Let  $X, Y : \Omega \rightarrow [0, 2]$  denote the projections  $X(x, y) = x, Y(x, y) = y$  and let  $T : \Omega \rightarrow \{0, 1\}$  be the map  $T(x, y) = \mathbf{1}_{[0, 1]^2}((x, y))$ . We will show that  $X, Y$  are conditionally independent given  $T$  but are not independent. First, note that for  $U \subset \Omega$  we have that

$$\mu(U) = \frac{1}{2}\lambda_2(U)$$

where  $\lambda_d$  denotes the Lebesgue measure on  $\mathbb{R}^d$ . It now follows that

$$\mu(U|T = 0) = \lambda_2(U \cap [0, 1]^2)$$

and

$$\mu(U|T = 1) = \lambda_2(U \cap [1, 2]^2).$$

Now for  $A, B \subset [0, 2]$  we have that

$$\begin{aligned} \mu(X^{-1}A \cap Y^{-1}B|T = 0) &= \lambda_2(A \times B \cap [0, 1]^2) \\ &= \lambda_1(A \cap [0, 1])\lambda_1(B \cap [0, 1]) \\ &= \lambda_2(X^{-1}A \cap [0, 1]^2)\lambda_2(Y^{-1}B \cap [0, 1]^2) \\ &= \mu(X^{-1}A|T = 0)\mu(Y^{-1}B|T = 0) \end{aligned}$$

and a similar calculation holds for conditioning on  $T = 1$ , which shows conditional independence.

Independence does not since if we take  $A = [0, 1], B = [1, 2]$  then

$$\mu(X^{-1}A \times Y^{-1}B) = \mu([0, 1]^2 \cap [1, 2]^2) = 0$$

but

$$\mu(X^{-1}A) = \mu([0, 1]^2) = \frac{1}{2}$$

and

$$\mu(Y^{-1}B) = \mu([1, 2]^2) = \frac{1}{2}.$$

**Example D.3** (Independent but not conditionally). Let  $\Omega = [0, 1]^2$  and suppose  $\mu$  is the uniform measure on  $\Omega$ . Let  $X, Y : \Omega \rightarrow [0, 1]$  be the projections. Then  $X$  and  $Y$  are independent. Let  $T : \Omega \rightarrow [0, 2]$  by  $T(x, y) = x + y$ . Then in fact  $X$  and  $Y$  are not conditionally independent given  $T$ . For example, let  $A = B = [0, 1/2]$ . Then for  $1 < t < \frac{3}{2}$  we have that  $(A \times B) \cap T^{-1}(t) = \emptyset$  and so  $P(A \times B|T = t) = 0$ . However  $P(X^{-1}A|T = t) > 0$  and  $P(Y^{-1}B|T = t) > 0$ .

Observe that we have a map  $S : \Omega \rightarrow \mathcal{Y} \times \mathcal{T}$  given by  $S(\omega) = (Y(\omega), T(\omega))$ . We let

$$\mu(A|Y = y, T = t) = \mu(A|S = (y, t)).$$

**Lemma D.4.** We have that  $X$  and  $Y$  are conditionally independent given  $T$  if and only if for all  $A \subset \mathcal{X}$  we have that

$$\mu(X^{-1}A|Y = Y\omega, T = T\omega) = \mu(X^{-1}A|T = T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega.$$

*Proof.* First assume  $X$  and  $Y$  are conditionally independent. Thus we must show that the right hand side satisfies the definition of conditional probability with respect to the random variable  $S$ . The right hand side is  $S$  measurable, as required. Thus it remains to show that

$$\int_{S^{-1}U} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) = \mu(X^{-1}A \cap S^{-1}U)$$

for all  $U \subset \mathcal{Y} \times \mathcal{T}$  and  $A \subset \mathcal{X}$ . By the monotone class lemma, it is enough to show it for  $U = B \times C$ . We have

$$\begin{aligned} \int_{S^{-1}(B \times C)} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) &= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{\Omega} \mathbb{1}_{Y^{-1}B}(\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{\Omega} E(\mathbb{1}_{Y^{-1}B} \mid T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{T^{-1}C} \mu(Y^{-1}B \mid T = T\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{T^{-1}C} \mu(Y^{-1}B \cap X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \mu(Y^{-1}B \cap X^{-1}A \cap T^{-1}C) \\ &= \mu(X^{-1}A \cap S^{-1}(B \times C)) \end{aligned}$$

as required.

Now we prove the converse. Thus we must assume that for  $A \subset \mathcal{X}$  we have

$$\mu(X^{-1}A \mid Y = Y\omega, T = T\omega) = \mu(X^{-1}A \mid T = T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega$$

and show that

$$\int_C \mu(X^{-1}A \mid T = t) \mu(Y^{-1}B \mid T = t) dT^* \mu(t) = \int_C \mu(X^{-1}A \cap Y^{-1}B \mid T = t) dT^* \mu(t)$$

for all  $C \subset \mathcal{T}$ . Using the law of total probability we have

$$\begin{aligned} \int_C \mu(X^{-1}A \cap Y^{-1}B \mid T = t) dT^* \mu(t) &= \mu(X^{-1}A \cap Y^{-1}B \cap T^{-1}C) \\ &= \mu(X^{-1}A \cap S^{-1}(B \times C)) \\ &= \int_{B \times C} \mu(X^{-1}A \mid Y = y, T = t) d(S^* \mu)(y, t) \\ &= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid Y = Y\omega, T = T\omega) d\mu(\omega) \\ &= \int_{Y^{-1}B \cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{\Omega} \mathbb{1}_{Y^{-1}B}(\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{\Omega} E(\mathbb{1}_{Y^{-1}B} \mid T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_{\Omega} \mu(Y^{-1}B \mid T = T\omega) \mathbb{1}_{T^{-1}C}(\omega) \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\ &= \int_C \mu(Y^{-1}B \mid T = t) \mu(X^{-1}A \mid T = t) dT^* \mu(t) \end{aligned}$$

□

## APPENDIX E. USEFUL RESULTS

**Lemma E.1.** Suppose that  $\nu$  is a  $\sigma$ -finite positive measure on  $X$ . Then there exists a probability measure  $\nu'$  such that  $\nu \ll \nu'$  and  $\nu' \ll \nu$  (i.e.,  $\nu(A) = 0$  if and only if  $\nu'(A) = 0$ ).

*Proof.* We only need to consider the case  $\nu(X) = \infty$ . There exists a countable partition  $X_1, \dots$  of  $X$  such that  $0 < \nu(X_i) = d_i < \infty$ . Now define

$$\nu'(A) = \sum_{i=1}^{\infty} 2^{-i} d_i^{-1} \nu(A \cap X_i).$$

□

**Lemma E.2.** Let  $\nu$  be a  $\sigma$ -finite positive measure on  $X$  and let  $\mathcal{P}$  be a family of probability measures on  $X$  such that  $P \ll \nu$  for all  $P \in \mathcal{P}$ . Then there exists  $a_1, a_2, \dots \in [0, 1]$  with  $\sum_i a_i = 1$  and  $P_1, P_2, \dots \in \mathcal{P}$  such that

$$P \ll \sum_{i=1}^{\infty} a_i P_i \quad \text{for all } P \in \mathcal{P}.$$

*Proof.* By the previous Lemma, we only need to focus on the case that  $\nu(X) = 1$ . Let

$$\mathcal{Q} = \left\{ \sum_{i=1}^{\infty} a_i P_i \mid \sum_i a_i = 1, a_i \geq 0, P_i \in \mathcal{P} \right\}$$

be the set of countable convex combinations of measures in  $\mathcal{P}$ . Observe that  $Q \ll \nu$  for all  $Q \in \mathcal{Q}$ . For  $Q \in \mathcal{Q}$  let  $f_Q = dQ/d\nu$  be the density of  $Q$  and let

$$S_Q = \{x \in X \mid f_Q(x) > 0\}.$$

We remark that  $f_Q$  and  $S_Q$  are almost surely well defined. Now let  $c = \sup_{Q \in \mathcal{Q}} \nu(S_Q) \leq 1$ . Let  $Q_1, Q_2, \dots \in \mathcal{Q}$  be such that  $\nu(S_{Q_i}) \rightarrow c$ . Now there exists  $Q_0$  such that

$$S_{Q_0} = \bigcup_{i=1}^{\infty} S_{Q_i}$$

and thus  $\nu(S_{Q_0}) = c$ . We claim that  $P \ll Q_0$  for all  $P \in \mathcal{P}$ , thus completing the proof. Suppose it were not the case. Then there exists  $A \subset X$  such that  $P(A) > 0$  but  $Q_0(A) = 0$ . Now let

$$Q' = \frac{1}{2} (Q_0 + P).$$

We have that  $Q_0 \ll Q'$  and so  $S_{Q_0} \subset S_{Q'}$ . Since  $Q_0(A) = 0$  we have that  $\nu(A \cap S_{Q_0}) = 0$ . On the other hand

$$0 < P(A) = \int_A f_P d\nu \leq 2 \int_A f_{Q'} d\nu$$

and so this means that there is  $A' \subset A$  with  $\nu(A') > 0$  and  $f_{Q'}(x) > 0$  for  $x \in A'$ . Thus

$$S_{Q_0} \cup A' \subset S_{Q'}$$

and so

$$\nu(S_{Q'}) \geq \nu(S_{Q_0}) + \nu(A') > c,$$

which is a contradiction. □